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Reliance on Schemas in Source Memory: Age Differences and Similarity of Schemas

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ABSTRACT

The present study has examined the effects of source-schema-consistency, schematic distinctiveness and encoding time of schematic information on source memory of younger and older adults. Participants were administered a source memory test, in which two sources presented statements that were congruent with, incongruent with or irrelevant to sources' professions. Professions of sources were schematically similar or distinct, and they were revealed before or after the presentation of items. The results yielded that revelation of profession information before presentation of items enhanced source memory for incongruent items. Aging was associated with a general decrease in source monitoring processes. Participants who were assigned to similar-source condition and had their schemas activated before learning phase had higher source monitoring scores. Results are discussed in relation to source monitoring framework and aging of memory.

Keywords: Source memory; Aging and memory; Schema; Source monitoring; Knowledge and memory; Age differences.

INTRODUCTION

Source memory is recalling the source of information as a result of a complex decision-making mechanism. In their source-monitoring framework (SMF), Johnson, Hashtroudi, and Lindsay (1993) argue that making source judgments partly relies on the prior knowledge about the source, such as schemas, stereotypes and categorizations, heuristics and plausibility. Several studies in the past decade focused on the source attribution processes and

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factors leading to source memory errors. Bayen, Nakamura, Dupuis, and Yang (2000), Holtgraves, Srull, and Socall (1989), Sherman and Bessenoff (1999), and Stangor (1988) have all shown that content of the statements and the schemas associated with the sources affect source memory decisions. Many errors in source attribution occur because a particular statement is more congruent with the schema of a gender, a social group, or a profession. As Bayen et al. (2000) suggested this may be due to a change in the strategy where participants make their guesses in congruence to schemas. Some other studies, on the other hand, showed that source monitoring for schema-inconsistent items may be equally as good as schema-consistent item monitoring (Hicks & Cockman, 2003; Lampinen, Copeland, & Neuschatz, 2001). This change in source memory performance is usually related to the time that prior knowledge about the source is made available to participants. That is, if prior knowledge about the source is made available before encoding, participants' source memory performance is significantly better, even for source monitoring of inconsistent statements.

The reliance on prior knowledge for biases in memory has received considerable attention in research on cognitive aging in the recent years. Memory performance was demonstrated to be particularly problematic for older adults in many studies, where older adults had less memory for schema-inconsistent items and more false alarms for non-presented schema-consistent items (Chen, 2004; Chen & Blanchard-Fields, 2000; Hess & Slaughter, 1991; Hess & Tate, 1991; Hess, Donley, & Vandermaas, 1989) Yet, reliance of older adults on prior knowledge for source monitoring has not been investigated exhaustively. One of the studies that has taken into consideration the age-related differences on how bias might operate in source monitoring was conducted by Mather, Johnson, and De Leonardis (1999). The study has shown that when statements uttered by sources were in agreement with the stereotypes about these sources, accuracy was higher. In addition, the statements which conflicted with the stereotype and agreed with the stereotype of the other source were more likely to be attributed incorrectly. The likelihood of incorrect attribution in the latter case was greater for older adults than younger adults.

One of the factors that might affect the biases of older adults for source monitoring may be related to the time that the prior information about the source is made available to the participants. Earlier studies yield conflicting results in relation to the availability of the source information beforehand. For example, Chen (2004) found that if older adults were provided with information about the sources before the learning phase, their context-induced biases were lower as compared to those participants provided with the same information at retrieval. One explanation for this finding might be that since participants know the schema related to the source before the learning phase, they might be able to bind item information to the source better than they would if they were revealed the schema information in the end, leading to improved source memory or better source guessing for the unexpected sources. On the other hand, another possibility for the availability of source information before the learning phase is that older adults may also have a tendency to ignore the unexpected information, leading to decreased source memory or decreased guessing for inconsistent items. Another study by Hess and Tate (1991), revealed that if older adults were provided information about the source before the study, they did not elaborate on the source information as much as younger adults did, leading to decreased source recognition. Therefore, in relation to the earlier studies, if prior knowledge related to the sources is made available to the participants before the learning phase, older adults may act in two different ways. One of them is that since they know the schema related to the source beforehand, they might be able to bind item information to the source better than they would if they were revealed the schema information in the end, leading to improved source memory or better source guessing for the unexpected sources. The other way is that they might ignore the unexpected information, leading to decreased source memory or decreased guessing for inconsistent items.

The primary purpose of this study is to investigate schema reliance of older adults using the methodology and the analyses employed by Bayen et al. (2000) and Hicks and Cockman (2003). In their study, Bayen et al. (2000) investigated the reliance of young adults on schemas where they manipulated the profession of the source. Two sources, Jim and Tom uttered statements that were doctor-consistent, lawyer-consistent or equally plausible for both professions and Jim's (doctor) and Tom's (lawyer) professions were told to the participants during retrieval. The results indicated that doctor-consistent statements were attributed to the doctor source more frequently and lawyer-consistent statements were attributed to the lawyer more frequently. In a follow-up study, Hicks and Cockman (2003) presented the profession information either before encoding or before retrieval using the same material. The results showed that the participants who received the profession information before encoding were more accurate in source identification for both schema-consistent and schema-inconsistent items than those participants who received profession information before retrieval.

In the present study, we have employed two age groups, as younger adults and older adults, in addition to the methodology of Bayen et al. (2000) and Hicks and Cockman (2003) to investigate whether older adults rely on schemas more than younger adults in making source memory decisions, especially in relation to the time that the prior knowledge about the source is revealed. It would be useful to understand the source attribution processes of older adults in the context of daily life examples. For example, an older adult might be more inclined to remember who recommended a particular vitamin in the country club if they knew that the person who recommended was a doctor rather than a lawyer. In both Bayen et al. (2000) and Hicks and Cockman (2003), the researchers studied the effects of schematic knowledge on source memory decisions in younger adults. Although Mather et al. (1999) have shown that stereotypes influence decisions by older adults considerably more, we focused on whether this influence pointed to a decline specific to source memory that was beyond what is accounted for by a general decline in cognitive functions. We have focused on the time that the schema was made available to the participants to explore how the availability of the schemas at learning phase might interact with the consistency of the information with the source on source memory performance of older adults. In addition, we employed two-high threshold multinomial modeling that disentangles item memory, source memory and guessing better than corrected measures of source memory (for a review, see Bayen, Murnane, & Erdfelder, 1996). In this study, we hypothesized that there would be a main effect for source memory performance with older adults relying on their schemas more than younger adults. However, we also expected that older adults may be able to make more strategic guesses if their schemas were activated before the learning phase, leading to increased source guesses for the unexpected statements than those older adults for whom the schemas are activated at retrieval.

In addition, we have added a manipulation of schematic similarity in order to explore how schematic similarity of sources may alter reliance on schemas in relation to aging. Studies examining the effects of distinctiveness have done so by employing perceptual (Ferguson, Hashtroudi, & Johnson, 1992; Johnson, De Leonardis, Hashtroudi, & Ferguson, 1995), temporal (Bayen & Murnane, 1996), or spatial (Ferguson et al., 1992) characteristics of the sources as markers of distinctiveness. Most of these studies provide evidence for better source memory in the conditions of highly dissimilar sources. However, in practice, source memory becomes more critical when one has to distinguish between two sources that have similar professional schemas. It becomes more critical at times, to distinguish between whether an utterance was said by an attorney or a judge, a doctor or a nurse, or a professor or an assistant. The schemas about these professions facilitate deducing plausible utterances, but similarity between the schemas may also become an impediment to guessing correctly. Thus, similarity in the form of schemas for the professions of the sources is an altogether different concept from the other forms of similarity employed in source memory studies because schemas are cues that are interlaced with the utterances whereas tone of voice, physical appearance, temporal or spatial positions are totally independent cues. Schema similarity may influence source memory in two opposite directions. Similar schemas for sources may lead to greater confusion and thus reduce source memory and lead to increased source guessing in general. However, similar schemas may also lead to better encoding since the participants elaborate on the source more in order to remember the items.

Therefore, in the present study we presented the participants with two sources, uttering statements which were schema-congruent (e.g., doctor source uttering doctor-consistent statements), schema-incongruent (e.g., doctor source uttering bank-teller consistent statements) or neutral (e.g., doctor uttering statements that are not relevant to her profession), in line with the original study of Bayen et al. (2000). The general design of the study was 2 (Age of participant: Young or Old) × 2 (Source distinctiveness: Doctor/Nurse or Doctor/Teller) × 3 (Source congruence: source-schema congruent statements, source-schema incongruent statements, source-schema neutral statements) × 2 (Schema activation time: Before or After the presentation of items).

An important aim of the current study was to investigate source monitoring differences from an aging perspective. The tests were administered to young and old adults to observe whether aging affects item and source memory in the same way. We have hypothesized that older adults' source memory performance would be lower than young adults' source memory, since earlier studies have found that there is a disproportionate decline in source memory for older adults. As supporting the earlier studies, older adults might also have a disproportionate level of source memory for congruent or incongruent items. That is, they might be inclined to have better source performance for items which are congruent with their sources, such as a doctor statement said by a doctor rather than a doctor statement said by a bank teller.

Secondly, there was a need to understand how schema information was used by older and younger participants in relation to the time that the schema was revealed to the participants. Therefore, the professions of the sources were revealed to the participants either before the presentation of the statements, allowing the participants to use the schema information during encoding, or they were revealed immediately following the statements, making this information available only for retrieval, as in Hicks and Cockman (2003). We hypothesized that older adults might have either have an advantage to remember the sources better, as in Chen (2004), or might be at a disadvantage, especially hindering them from monitoring sources for schema-inconsistent items, as in Hess and Tate (1991).

Finally, in order to understand the effects of source similarity, the participants were presented either two sources who would be considered schematically similar, Doctor and Nurse, or two sources who are very distinct from each other, Doctor and Bank Teller. These sources uttered to participants statements that were (a) congruent with their schema (e.g., doctor source uttering typically doctor statements), (b) incongruent with their schema (e.g., bank teller uttering typically doctor statements), and (c) neutral statements which could be uttered by anyone regardless of profession. The earlier studies have suggested that older adults might have decreased levels of source memory performance if the sources are similar to each other. However, the similarity of the sources was through the use of physical cues such as tone of voice, appearance, temporal or spatial position rather than schematic knowledge interlaced with the profession of the source. Therefore, we hypothesized that distinctive cues may not operate the same way as it does in independent physical characteristics.

One of the modifications different from Bayen et al. (2000) and Hicks and Cockman (2003) studies was that we employed distracters that were not only lexically but also semantically different. Another important difference of the current study from Bayen et al. (2000) and Hicks and Cockman (2003) was manipulation of schematic distinctiveness of sources by pairing a doctor source with either a nurse source or a bank teller source.

METHOD

Participants

The sample comprised of 80 participants, 40 younger and 40 older adults. Forty young adults from Koç University, consisting of 22 males and 18 females, participated in the research voluntarily, and received extra credit from an introductory course in Psychology. They had a mean age of 19.53 with standard deviation of 1.15 (min = 17, max = 22). They had a mean education of 13.33 years (min = 11, max = 16).

Older adults were community-dwelling adults in Istanbul. The older sample consisted of 24 females and 16 males, who participated in the research voluntarily. The older adults had a mean age of 67.13 (*SD* = 6.17, min = 60, max = 82). The participants were included in the sample on conditions that they were at least high school graduates, had no major health problems, and had a total of more than 25 points out of 30 in Mini Mental State Examination (MMSE). Older adults who had serious health problems or a history of brain trauma, stroke and dementia, were not included in the study. The older participants had a mean education of 13.88 years (min=11, max = 19).

There were no significant differences between young and old adults in terms of years of education. Self health ratings revealed no significant differences for older adults and younger adults.

Materials

Mini Mental State Examination (MMSE)

Mini Mental State Examination (Folstein, Folstein, & McHugh, 1975) is one of the most widely used screening tests for frontal lobe dysfunctions, especially in older adults. It has been tested in many studies for validity and reliability (e.g., Tombaugh, 2005).

MMSE was translated into Turkish and was used as a screening test before participation in the study. The one-way ANOVA showed that MMSE score was significantly associated with age ($F(1, 79) = 9.72$, $p < .01$, $MSE = 1.32$), younger adults ($M = 29.23$, $SD = .73$) scoring significantly higher than older adults ($M = 28.43$, $SD = 1.45$). Even though this difference was statistically significant, the older adults had rather high scores, which is consistent with the literature. For example, in a Canadian sample of 7754 older adults, older adults between the ages of 65 and 69 had a mean of 27.7 with a standard deviation of 2.5 (Bravo & Hébert, 1997).

Source Memory Test

Four different groups of items were generated, containing 189 statements. Groups of items contained (a) typical statements that a doctor would say, (b) typical statements that a nurse would say, (c) typical statements that a bank teller would say, and (d) statements that are equally plausible for all these professions (neutral statements). A pool of 10 participants, consisting of nurses, bank workers, psychology students and other adults, were asked to rate these sentences twice. In the first rating, they were asked to rate these 189 sentences in terms of their distinctiveness as doctor-relevant and nurse-relevant statements by a semantic differential method. The participants could rate the sentences as uttered more often by doctors, uttered more often by nurses, equidistant to both professions or as not related to any of these professions. In the second rating, the same procedure was repeated using bank teller instead of nurse in the instructions. The final list of items that were classified as 'statements distinctly belonging to doctors', 'statements distinctly belonging to nurses', 'statements distinctly belonging to bank tellers' and 'statements that are plausible for all professions' (22 items for each group) were used as target or distracter items. The neutral sentences were chosen by selecting those sentences that were rated as equally plausible for all professions. All statements were assured to be semantically different.

Two separate PowerPoint slide shows were prepared for each distinctiveness condition. Slides consisted of either a combination of doctor, nurse and neutral statements (low distinctiveness condition) or a combination of doctor, bank teller and neutral statements (high distinctiveness condition). There was a total of 48 items randomly assigned to two different female sources with typical Turkish names. Each of the sources uttered 24 items, comprised of eight source-congruent, eight source-incongruent and eight neutral items. Photographs of two women with distinctive facial features about ages 30–35 and two female voices supplemented the presented items. While participants heard the sources uttering the sentences, they could also read what was being said, printed in white uppercase letters on a dark blue background, alongside the pictures. The names of the sources were also printed under the photographs on each slide. The presentation order of the slides was first randomized and then adjusted for balanced presentation according to source and source-congruence of the statement, such that there were equal number of slides for each source uttering source-congruent, source-incongruent, and neutral items, in the beginning, in the middle, and at the end of the presentation. The duration for the presentation of the slides was determined in a pilot study, where three old adults and three young adults were asked to view the items in a self-paced manner. The average duration obtained was used in the main study such that each item was presented at the same pace of 7 s.

The profession information was given either at the beginning or at the end of the learning phase. One of the pictured women was identified as the doctor in both high and low distinctiveness conditions. The other face was identified as the 'nurse' in the low distinctiveness condition and as 'bank teller' in the high distinctiveness condition.

The source memory test, administered on the same computer, immediately after the presentation, contained 36 target statements from the learning phase. In order to prevent primacy and recency effect, 12 of the items in the learning phase (six items presented at the beginning of the learning phase and six items presented at the end of the learning phase) were excluded. Additionally, 18 distracter items, containing six doctor-consistent statements, six other profession-consistent statements and six neutral statements were included in the recognition test, adding up to a total of 54 statements for the test phase. The statements in the test phase were printed in white on a dark blue background and were all articulated by a third female voice. Below the statements, photographs of the two sources and the word 'NEW' were shown to the participants instructing them to make a choice among the three options. The order of presentation was randomized for the test phase as well.

Procedure

The experimental sessions were conducted individually either at a quiet room of the university or at the residence of the participant. The participants were told that they were going to have a test of memory, consisting of two parts. In the first part, the experimenter administered MMSE to the participants orally.

The second part was conducted on a laptop computer. After participants were randomly assigned to one of the four conditions (highly distinctive sources or similar sources and profession information provided before or after the presentation of statements), they were told that they would watch a slide show, in which two sources uttered statements. Participants were informed that they would see pictures of two women, making statements, accompanied by their voices, uttering the written material on the slide. Participants were instructed to watch the slides and listen carefully because they would later be tested for their memory. No further information about the test was provided. Depending upon the experimental condition, participants were informed about the profession of the source either before or after the learning phase by an additional slide.

After the learning phase was over, participants were given the instructions for source memory test. Participants were told that they would be shown statements on the screen, which were either said by one of the two sources during the learning phase or statements that were not heard during the presentation at all. The participants were instructed to identify whether the item was said by source 1, source 2, or neither of them.

RESULTS

Several computations were made on responses of participants for different types of analyses. These included conditional source identification scores (CSIM) and average conditional source identification measure (ACSIM) for eliminating guessing bias in source memory scores and two-high threshold multinomial modeling of source memory, which gives unconfounded measures of item memory, source memory and guessing bias (Bayen et al., 1996).

Simple comparisons and interaction contrasts are significant at the alpha level of .5, unless otherwise stated.

Conditional Source Identification and Average Conditional Source Identification Measures

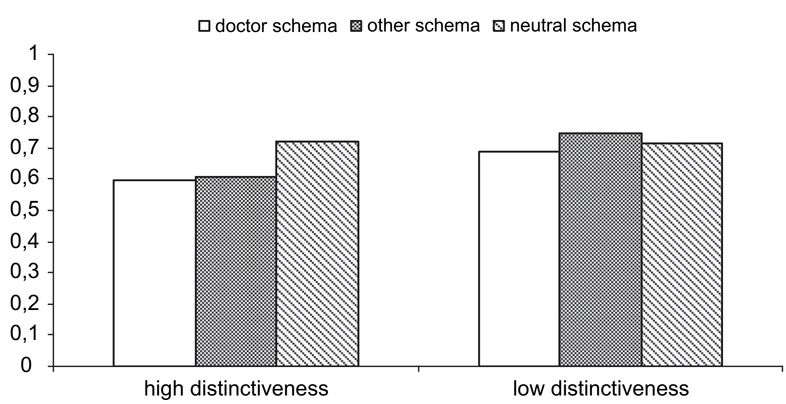
CSIM is a method for analyzing hit rates in source monitoring taking into account how response bias may operate in guessing between two old sources. CSIM is calculated by dividing hits for a source by the sum of correct attributions and misattributions for that source. For example, a participant's 'doctor' responses to doctor statements uttered by the doctor source were divided by the sum of responses as 'doctor' and 'other-profession' responses for doctor statements uttered by the doctor. In this case, if the participant has relied on schemas by attributing all the doctor-consistent items to the doctor, the score would be at the chance level of 50%, because both sources uttered an equal number of doctor-consistent items. Thus, a response rate over .5 is considered to be above chance level and indicates a better source memory, whereas response levels equal to or below .5 indicates guessing rather than remembering the source of the statements. A disadvantage of CSIM is that these scores do not take into consideration the effect of any item that was defined as 'new'; therefore, they may not be able to measure pure source-identification. However, they are confirmed to measure source memory better than corrected measures of hits (Murnane & Bayen, 1996).

We have conducted a mixed-subjects analysis of variance (ANOVA) with age (young vs. old), distinctiveness (distinctive vs. similar sources), schema activation time (before the learning phase vs. after the learning phase) as the between subjects factors and item type (doctor-consistent, other profession consistent, and neutral items) and source (source A and source B) as the within subjects factor on CSIM scores.

As we have hypothesized, the main effect of age was significant for CSIM scores ($F(1, 71) = 12.774$, $p < .01$, $MSE = .096$), with younger adults having better source memory than older adults. It is important to emphasize that older adults had higher scores than .5, indicating that they had some source memory ($M = .620$, $SD = .320$); however, younger adults had significantly higher scores than older adults, indicating their advantage for memory ($M = .730$, $SD = .279$). The age of the participants did not interact with other variables, showing that reliance on schemas was not differential for older and younger adults in relation to activation time and schema distinctiveness.

The main effect of item type was significant ($F(2, 142) = 6.313$, $p < .01$, $MSE = .035$). Pairwise comparisons revealed that the sources for neutral items ($M = .716$, $SD = .266$) were identified better than doctor-consistent items ($M = .642$, $SD = .342$). The other-profession-consistent items ($M = .678$, $SD = .316$) were not significantly different than either doctor or neutral statements. The main effect of distinctiveness was significant ($F(1, 71) = 7.309$, $p < .01$, $MSE = .096$). Participants assigned to low distinctiveness group had higher CSIM scores ($M = .717$, $SD = .176$) as compared to participants in the high distinctiveness group ($M = .640$, $SD = .258$). This was overridden by an interaction between item type and distinctiveness ($F(2, 142) = 6.383$, $p < .01$, $MSE = .035$). As shown in Figure 1, the pairwise comparisons showed that participants in the low distinctiveness condition attributed both

FIGURE 1. Interaction of distinctiveness and schema for CSIM scores.

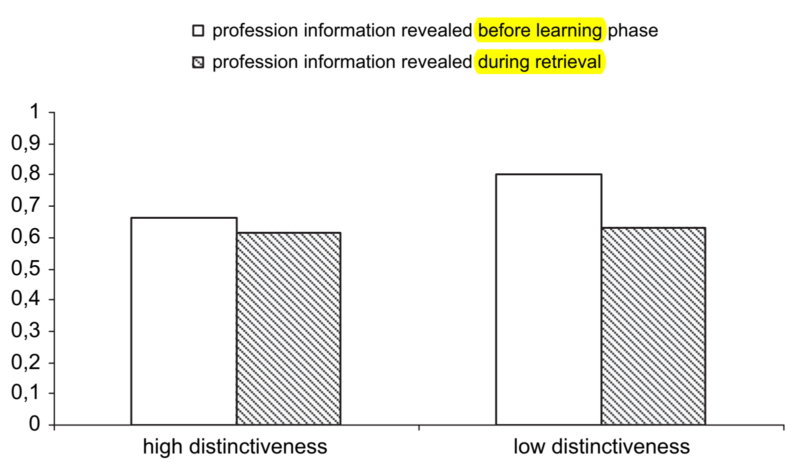


doctor-consistent statements (M = .690, SD = .337) and other-profession-consistent statements (M = .749, SD = .179) correctly to their sources more often than participants in high distinctiveness condition, whereas CSIM scores for attribution of neutral statements did not differ between high (M = .719, SD = .261) and low distinctiveness (M = .713, SD = .287) conditions. Moreover, CSIM scores for doctor-consistent, other-profession-consistent and neutral statements were correctly attributed comparable in low distinctiveness condition, whereas CSIM scores for neutral statements (M = .719, SD = .261) were higher than doctor-consistent (M = .594, SD = .321) and other-profession-consistent statements in high distinctiveness condition (M = .607, SD = .343).

The main effect of schema activation time was significant as well ($F(1, 71) = 14.925$, $p < .001$, $MSE = .096$). Participants who were given schema-activating information before the presentation had higher CSIM scores (M = .734, SD = .243) than participants who were given the same information after the presentation was finished (M = .623, SD = .254). This was qualified by a significant interaction between distinctiveness and schema activation time ($F(1, 71) = 4.330$, $p < .05$). Simple comparisons revealed a marginally significant difference between means. As shown in Figure 2, participants in low distinctiveness group who were given profession information before had higher CSIM scores than all other groups. The time the profession information was given did not affect CSIM for participants in high distinctiveness group, whereas being given profession information before was associated with higher CSIM scores in low distinctiveness group.

We have also replicated the finding of Hicks and Cockman (2003). The three-way interaction of source, item type and schema activation time was

FIGURE 2. Interaction of distinctiveness and schema activation time for CSIM scores.



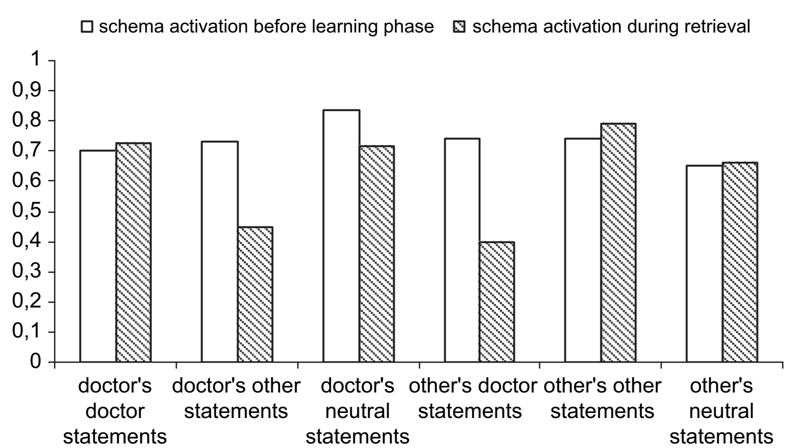
significant ($F(2, 142) = 8.890$, $p < .001$). Simple effect comparisons showed that when participants were given profession information before the presentation of items, the performance for all kinds of statements by both sources were comparable (see Figure 3). On the other hand, when participants were given profession information after the presentation, their performance decreased significantly for doctor-consistent items that the other source uttered ($M = .398$, $SD = .369$) and the other-profession-consistent items that doctor uttered ($M = .448$, $SD = .374$).

In addition, average conditional source identification measure (ACSIM) was calculated by taking the average of the CSIM scores separately for congruent, incongruent and neutral items. ACSIM for expectancy of statements was calculated by taking the average of the CSIM scores for all cells. For example, ACSIM for neutral sentences were calculated by taking the average CSIM score of doctor's neutral statements and nurse's neutral statements. In the end, three different ACSIM scores were calculated for each participant as source-schema-congruent, source-schema-incongruent and neutral. A mixed ANOVA was conducted using ACSIM scores as the within-subjects variable and age, distinctiveness and schema activation time as between-subjects variables. ACSIM scores provided further evidence for our hypotheses.

The main effect of age was significant ($F(1, 71) = 12.774$, $p < .01$, $MSE = .048$). Older adults had lower ACSIM scores ($M = .628$, $SD = .265$), as compared to younger adults ($M = .730$, $SD = .232$).

The main effect of distinctiveness was significant ($F(1, 71) = 7.309$, $p < .01$, $MSE = .048$). Participants in the low distinctiveness condition had higher ACSIM scores ($M = .717$, $SD = .176$) as compared to participants in the high distinctiveness group ($M = .640$, $SD = .258$).

FIGURE 3. Interaction of source, schema and schema activation time for CSIM scores.



The main effect of source-schema-congruence ($F(2, 142) = 9.318$, $p < .001$, $MSE = .064$) and schema activation time ($F(1, 71) = 14.925$, $p < .001$, $MSE = .048$) were significant. These main effects were further qualified by a significant interaction between source-schema-congruence and schema activation time ($F(2, 142) = 10.115$, $p < .001$, $MSE = .048$). Planned comparisons revealed that ACSIM score for source-schema incongruent statements ($M = .735$, $SD = .738$) were significantly higher when profession information was given before the presentation. Moreover, source-schema-incongruent statements ($M = .423$, $SD = .345$) were significantly remembered less as compared to source-schema-congruent ($M = .759$, $SD = .236$) and neutral sentences ($M = .688$, $SD = .186$) when the profession information was released after the presentation.

Two-High Threshold Multinomial Modeling Results

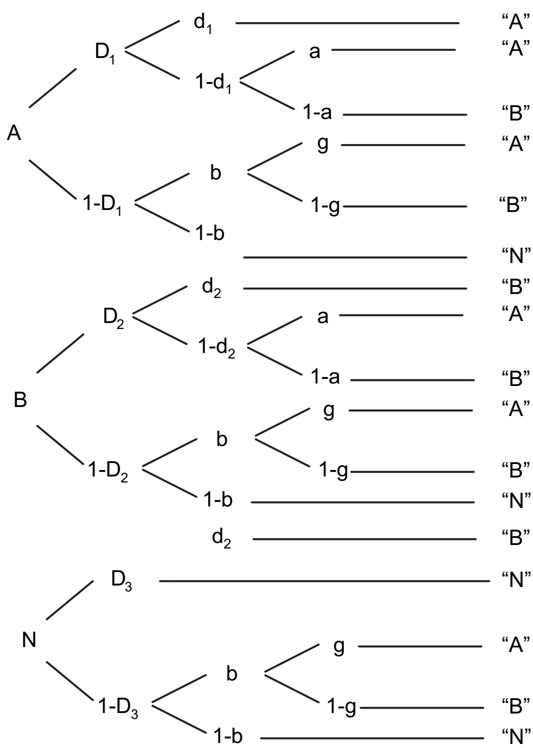
We have used two-high threshold multinomial modeling to disentangle the item memory, source memory and guessing (Bayen et al., 1996). The parameters and their explanations are shown in Figure 4.

This model has the distinct advantage of providing unconfounded pure measures of item recognition, source memory and guessing biases. We have used submodel 4 of the 2HT multinomial model since it is the most parsimonious model (Bayen et al., 1996). We have arbitrarily chosen the doctor source as Source A. We have made the following restrictions as (1) item recognition is equal for doctor source, the other source and new items ($D_1 = D_2 = D_3$), (2) source memory is equal for items presented by the doctor and by the other source ($d_1 = d_2$), and (3) probability of guessing ‘doctor source’ and ‘other source’ is equal for recognized and unrecognized items (a = g). Therefore, the four parameters of the model are D, the probability an item is recognized; d, the probability that source for the item is correctly recognized; b, probability of guessing that an item is old; and g, the probability of guessing that an object is presented by the doctor source.

We have chosen a confidence interval of 99% given the large sample size, since even small deviations between data and model predictions (W=0.1) would be detected almost with certainty ($(1-\beta) > .999$) and analyzed the data. The model provided a good fit [G^2 (48) = 72.5) in the overall at a confidence interval of 99%. Parameter estimates of the data for high distinctiveness condition and low distinctiveness condition are presented separately in Tables 1 and 2, respectively.

We have conducted significance tests for item memory parameter D, source memory parameter d and guessing bias parameter g to determine whether there were significant differences between old and young adults as well as between the schema activation times as before the learning phase or as at retrieval. These tests indicated that for both young adults [G^2 (2) = 7.61) and older adults [G^2 (2) = 18.33), the sources for doctor expected

Figure 4. Two-high-threshold multinomial model of source monitoring.
Two-high-threshold multinomial model of source monitoring. A = Source A item; B = Source B item; N = Distracter item; D_1 = probability of detecting a source an item source A; D_2 = probability of detecting a source an item source B; D_3 = probability of detecting that a distracter is new; d_1 = probability of correctly discriminating the source of an item is from Source A; d_2 = probability of correctly discriminating the source of an item is from Source B; a = probability of guessing that a detected item is from Source A; b = probability of guessing that an item is old; probability of guessing that an undetected item is from Source A (taken from Bayen, Murnane and Erdfelder, 1996).



statements said by the doctor source were significantly better remembered when profession information was presented before the learning phase as compared to when profession information was revealed at retrieval.

Analyses of multinomial-based parameters revealed that there were no differences in item recognition (parameter D), source memory (parameter d) and guessing bias (parameter g) for comparisons between age groups and schema activation time.

DISCUSSION

The current study was conducted to explore how schemas affect source monitoring and guessing biases in older and younger adults. One of the main

TABLE 1. Parameter Estimates, Confidence Intervals for the Four-parameter Two-High Threshold Multinomial Model of Source Monitoring for Young and Old Adults in High Distinctiveness

		Doctor Schema		Bank Teller Schema		Neutral Schema	
Schema		Time of Schema Activation		Time of Schema Activation		Time of Schema Activation	
Parameters	Age group	Before	After	Before	After	Before	After
D	young	.76 (.66-.85)	.75 (.65-.85)	.54 (.42-.66)	.53 (.41-.68)	.80 (.72-.88)	.69 (.58-.79)
	Old	.53 (.41-.65)	.63 (.51-.75)	.54 (.42-.66)	.56 (.44-.68)	.67 (57-.77)	.67 (.56-.77)
d	young	.24 (.04-.44)	.22 (.01-.42)	.43 (.16-.70)	.45 (.21-.68)	.63 (.46-.80)	.74 (.54-.94)
	Old	.24 (-.03-.50)	.00 (-.22-22)	.22 (-.04-.48)	.00 (-.24-.24)	.41 (.19-.63)	.21 (-.01-.48)
b	young	.34 (.13-.55)	.33 (.12-.54)	.40 (.25-.55)	.25 (.10-.38)	.17 (-.03-.36)	.37 (.19-.56)
	Old	.25 (.11-.39)	.50 (.33-.66)	.29 (.15-.43)	.33 (.19-.48)	.25 (.09-.43)	.25 (.08-.42)
g	young	.68 (.57-.80)	.66 (.55-.78)	.33 (.21-.46)	.19 (.06-.31)	.58 (.40-.77)	.30 (.13-.47)
	Old	.44 (.32-.57)	.73 (.64-.82)	.60 (.47-.72)	.26 (.17-.36)	.59 (.45-.73)	.59 (.47-.71)

Table 2. Parameter Estimates, Confidence Intervals for the Four-parameter Two-High Threshold Multinomial Model of Source Monitoring for Young and Old Adults in Low Distinctiveness

		Doctor Schema		Bank Teller Schema		Neutral Schema	
Schema		Time of Schema Activation		Time of Schema Activation		Time of Schema Activation	
Parameters	Age group	Before	After	Before	After	Before	After
D	young	.69 (.59-79)	.69 (.59-.79)	.75 (.65-.85)	.70 (.59-.81)	.70 (.60-.80)	.75 (.65-.85)
	old	.53 (.39-.66)	.44 (.30-.58)	.69 (.58-.81)	.48 (.34-.62)	.57 (.45-.70)	.60 (.48-.72)
d	young	.75 (.57-.94)	.17 (-.04-.39)	.89 (.72-1.06)	.40 (.19-.62)	.69 (.50-.87)	.55 (.36.75)
	old	.80 (49-1.11)	.20 (-.12-.52)	.84 (.61-1.05)	.48 (.17-.80)	.51 (.26-.77)	.27 (.03-.52)
b	young	.27 (.09-.44)	.37 (.19-.56)	.46 (.26-.67)	.50 (.32-.68)	.27 (.09-.45)	.33 (.13-.54)
	old	.60 (.47-.73)	.56 (.48-.69)	.64 (.48-.81)	.64 (.53-.77)	.35 (.20-.50)	.34 (.18-.50)
g	young	.44 (.24-.65)	.69 (.58-.79)	.26 (.07-.46)	.28 (.16-.40)	.70 (.52-.87)	.55 (.36-.75)
	old	.36 (.22-.50)	.70 (.60-.80)	.49 (.33-.66)	.23 (.12- .32)	.64 (.51-.79)	.64 (.52-.76)

research questions posed by the current study was whether age differences would yield differential results for older and younger adults. We had hypothesized that older adults may depend more on their schemas in order to compensate for the losses in their source memory. There was a main effect for age groups in conditional source identification scores of older and younger adults, confirming that source memory of older adults is more impaired than young adults. However, we failed to find any interactions between age group and item type, indicating that older adults may be using the similar strategies to monitor sources as young adults. Moreover, we have not found any significant results related to multinomial modeling of source memory for guessing bias (parameter g), providing further proof that older adults may not have differential reliance on schemas.

Another research question we posed was how activation of schematic knowledge may influence reliance on schemas. We posed the question of whether having prior knowledge about a source before the presentation of items would facilitate or inhibit source monitoring. The results indicated that participants had better memory for source if profession information was provided before the participants started encoding the items. Yet, the main effect of schema activation was further qualified by source-schema-congruence. Participants had comparable conditional source identification measure scores for congruent, incongruent and neutral items when profession information was released in advance.

This finding replicated that of Hicks and Cockman (2003). One of the assertions of SMF for successful source attributions is related to how well the source is encoded (Johnson et al., 1993). Johnson et al. (1993) contends that source memory depends upon the information available from activated memory records; therefore, it relies fundamentally on the quality of the initially encoded information. Anything that prevents a person from fully encoding source information at acquisition will reduce the possibility that source or context will be remembered. In this study, manipulating the availability of schematic information about source professions has led to results confirming SMF accounts of source memory. If participants were given profession information before encoding the items, they had increased levels of ACSIM scores for source-schema incongruent items, compared to those who were revealed the profession information after encoding the statements. Therefore, making profession information available at encoding may have helped participants contextualize information, especially in specifying the source of schema-incongruent items. On the other hand, adults who had no schematic information about source during acquisition of items could not contextualize information as much, leading to decreased ACSIM scores for incongruent items.

The current study also assessed the influence of the similarity between source schemas by exposing the participants to either two similar profession sources (doctor and nurse) or two distinctive profession sources (doctor and bank-teller). One of the interesting findings that we have obtained was related to the interaction between source similarity and schema activation time. The results indicated that source monitoring performance of participants assigned to low distinctiveness condition and given the profession information before the learning phase had better CSIM and ACSIM scores than all other groups. One possible reason for the current finding may be related to the use of cognitive resources. When participants were warned in advance that they would hear statements from a doctor and a nurse, their attention may have been significantly drawn towards distinguishing statements of nurse and doctors, since the statements were similar to each other, leading to increased elaboration for sources. Yet, when they were told that they would hear items from a bank teller and a doctor, they may have speculated that they would distinguish between the two sources more easily; therefore, they made less elaborations, leading to decreased hit rates. Even if they did not anticipate a demand to remember the source of statements, similarity of sources may have reduced confidence in the ability to recall by relying on schemas. Consequently, other strategies to encode items more specifically may have been at work. On the other hand, when participants were given profession information after the learning phase was over, they could remember the items, but since they had not encoded the items in the relation to the source of the item, leading to decreased ACSIM scores.

The multinomial modeling has revealed that the results for this increase originated from the increase in source memory for doctor-expected statements by the doctor source when profession information was revealed before the learning phase. To clarify, when schemas were activated before the learning phase in low distinctiveness condition, this enhanced the source memory for doctor's doctor-expected statements as compared to when schemas were activated at retrieval. Considering that the doctor statements used in both high and low distinctiveness condition are the same statements, this was clearly a context effect. That is, when doctor statements were presented in the context of doctor–nurse dyads, the source memory for the doctor-expected statements increased. It may be that participants paid more attention to distinguish doctor and nurse statements, leading to an increased source memory for especially congruent statements.

The results of the current study are in conflict with earlier studies employing source similarity as a variable (e.g., Bayen & Murnane, 1996; Ferguson et al., 1992; Johnson et al., 1995). The schematic similarity of the sources in the current study may have helped source monitoring rather than inhibiting the processes. A possible reason for the contrast between the results of the current study and previous ones may relate to the fact that former studies have not manipulated source similarity on the basis of source schemas. They have usually focused on perceptual, temporal, or spatial characteristics of sources rather than schematic similarity. Information about schematic similarity ahead of presentation may have instigated a different processing strategy than the strategies employed when sources appear with different perceptual characteristics. They engaged more cognitive resources and operations between nurse and doctor utterances, leading to better source recognition, whereas the greater distinction between the doctor and the bank-teller allowed for an assumed confidence and less cognitive effort.

To summarize, the current study replicated the findings of Hicks and Cockman (2003), regarding source-schema congruence and time of schema activation. The major findings were that the ability to identify the source increased when the item schema was congruent with source and decreased when the item schema was incongruent. Yet, these influences were further qualified by an interaction of source-schema congruence and schema-activation time. When schemas were activated before the presentation of items, all types of items were remembered at the same rate, regardless of source-schema-congruence. On the other hand, source memory for source-schema-incongruent items decreased if schemas were activated after the presentation of items. It was also observed that even though source memory was more impaired in older adults, there was no indication for any differential processing patterns than younger adults. This implies that older adults may be more likely to remember incorrectly that a doctor recommended a vitamin to them but the increased likelihood for this confusion would not be because older adults rely more on doctor schemas when making such judgments.

The level of distinctiveness was the variable in the current study that caused surprising results. Previous findings in the literature have strongly asserted that as similarity of sources increase, source memory suffers from similarity. Contrary to the source-monitoring framework and the earlier findings, the present study has found source similarity to enhance source identification at certain conditions. Similarity of statements and sources possibly instigated a strategy in processing. The participants in the low distinctiveness group who were given profession information beforehand may have encoded information more carefully because the sources were difficult to distinguish. The participants in the high distinctiveness group did not engage in any additional cognitive strategies because the sources availed themselves for easy discrimination. It should be noted, however, that while the level of schema distinctiveness was manipulated at the perceptual level, the sources in both distinctiveness conditions were equally distinct. This divergence from other studies regarding similarity is significant to note because it points to differential encoding strategies for perceptual characteristics which have no relevance to what was said, and for schematic information about sources which may interact with the content of the utterances.

One of the problems of the current study was that since the number of items was relatively small for conducting multinomial analyses, the results for multinomial analyses have large confidence intervals, which may have increased the type II error for rejecting the null hypotheses. Future studies should aim to use more items or more participants so that the results can be reported with lower confidence levels.

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